

**A PROPOSED OFFERING OF
A VILLAGE RESIDENT INFORMATION & PROFILING SYSTEM
FOR DISIPLINA VILLAGE, UGONG**

A Project Proposal Presented to the
Faculty of Datamex College of Saint Adeline, Inc.

In Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Information Technology

By:

Din, Samantha Rose

Factor, Carlo C.

Sto Domingo, Joshua

Submitted to:

Mr. Gabriel Thomas Torneros

TESTING

DOCUMENTATION

INTRODUCTION

The testing phase was carried out to ensure that the Village Resident Information & Profiling System functions correctly and serves its intended purpose. This stage helps verify that all parts of the system work as designed and that users can perform their tasks without issues. Testing also helps identify any errors or bugs that may affect system performance. It ensures that the system runs smoothly before it is used in real-world operations.

During the testing process, each feature of the system was checked carefully to confirm that it performs according to the requirements. This includes logging in, registering new residents, managing household information, and assigning roles between the admin and staff. Every action was tested to ensure that the system responds correctly and displays accurate information. The goal is to make sure that both admin and staff can use the system with ease and confidence.

Another focus of the testing was to evaluate how user-friendly the system is for both technical and non-technical users. The testers observed whether buttons, forms, and tables worked properly and if the layout was easy to navigate. Any confusing parts or unclear instructions were recorded for improvement. This helps make the system more intuitive and comfortable for users in the barangay office.

However, the testing phase did not include performance testing on servers or third-party system integration. Instead, it concentrated on the system's core features, such as data entry, record updates, and role-based access. These functions are essential for daily operations and were given the highest testing priority. Overall, the testing ensured that the system is reliable, efficient, and ready for actual use in managing village records

TESTING ENVIRONMENT

Hardware Specifications:

Testing was performed on a standard desktop setup suitable for most barangay offices.

- Processor: Intel Core i5 11th Gen (2.40 GHz)
- Memory: 8 GB RAM
- Storage: 256 GB SSD
- Operating System: Windows 10 Pro (64-bit)

Software Requirements:

- Programming Language: Visual Basic .NET (VB.NET)
- Database Management System: Microsoft SQL Server Management Studio (SSMS)
- Development Environment: Microsoft Visual Studio 2022
- Testing Type: Manual Testing using Form Inputs and Validation Checks
- Operating System: Windows 10 Pro (64-bit)

Test Data:

For testing, sample accounts and resident records were created to simulate real scenarios. Example data included:

- Resident: Juan Dela Cruz
- Household Head: Maria Santos
- Admin Account: test_admin / admin123
- Staff Account: test_staff / staff123

TESTING METHODOLOGY

The project used black-box testing and user acceptance testing (UAT) to ensure all features function correctly. Black-box testing focused on testing inputs and expected outputs without looking into the actual code. Meanwhile, user acceptance testing allowed actual users the admin and staff to test how the system works in real use. Each feature was reviewed for accuracy, ease of use, error handling, and responsiveness across different browsers and devices.

The testing criteria focused on ensuring that all inputs accept only valid data, user roles are properly restricted, and error messages appear when needed. It also made sure that data could be saved and retrieved correctly, without any loss or delays during operations.

TEST CASES

The **test cases** describe the different scenarios used to check if each part of the system works correctly. Each case includes the steps taken, the expected result, and the actual outcome after testing. This helps ensure that all system functions such as login, registration, and record updates perform as intended. Through these test cases, errors can be detected early and fixed before the system's final deployment.

Test Case ID	Test Description	Test Steps	Expected Output	Actual Output	Status	Remarks
TC001	Login with valid credentials	1. Enter username	User is redirected to dashboard	User is redirected to dashboard	Pass	Works properly
		2. Enter password				
TC002	Login with invalid password	3. Click login	Error message appears	Error message appears	Pass	N/A
		1. Enter username				
TC003	Register new resident	2. Enter wrong password	Confirmation message appears	Confirmation message appears	Pass	N/A
		3. Click login				
TC004	Approve	1. Fill in registration form	Resident	Resident	Pass	N/A
		2. Click submit				
		3. Wait for admin approval				

Test Case ID	Test Description	Test Steps	Expected Output	Actual Output	Status	Remarks
TC005	resident registration	logs in	status changes	status changes		
		2. Opens pending list	to approved	to approved		
		3. Click approve				
	Add household group	1. Admin clicks "Add Household"	Household group added successfully	Household group added successfully	Pass	N/A
TC006	Edit resident info	2. Enters household head name				
		3. Click save				
		1. Select resident	Resident details updated	Resident details updated	Pass	N/A
		2. Click edit				
TC007	Staff view restriction	3. Update information				
		4. Save changes				
		1. Login as staff	Edit button is disabled	Edit button is disabled	Pass	Restriction working
TC008	Logout function	2. Try to edit records				
		1. Click logout button	Redirected to login page	Redirected to login page	Pass	N/A

BUG TRACKING & ISSUE LOG

The bug tracking and issue log records all the problems or errors found during testing. Each bug is listed with its description, severity level, status, and resolution to help the team monitor progress. This log helps developers identify which issues need urgent fixes and which can be handled later. Keeping a clear record of bugs ensures the system remains stable, efficient, and error-free before release.

Bug ID	Description	Severity	Reported By	Status	Resolution
B001	Upload button fails when image exceeds 2MB	Medium	Tester 1	Resolved	Added file size limit and error alert
B002	Household table doesn't refresh after deleting a record	Low	Tester 2	In Progress	Will fix in next version
B003	Admin dashboard loads slowly with many records	Medium	Tester 3	Open	Optimize database queries
B004	Missing error alert for blank required fields	High	Tester 1	Resolved	Added validation messages

USER ACCEPTANCE TESTING (UAT)

During the UAT phase, both the admin and staff tested the system to see how well it supports their daily work in the barangay office. They explored each feature, such as login, resident registration, and household management, to ensure everything functioned smoothly. The testers found the overall design simple and easy to navigate, even for users with limited technical experience. This helped confirm that the system was practical and ready for real use.

The staff users shared that the system made their work faster and more organized compared to manual recording methods. They were able to check and verify resident information more efficiently, saving time during data entry and updates. Meanwhile, the admin users appreciated how easy it was to manage accounts, approve registrations, and generate reports. The system's layout and flow helped reduce confusion and improve accuracy in handling records.

CONCLUSION & RECOMMENDATIONS

Based on the overall test results, the Village Resident Information & Profiling System performed well and met its intended purpose. Most of the functions worked as expected, and only minor issues were found during testing. After fixes and adjustments, the system became stable and ready for deployment.

It is recommended to continue improving system performance, especially for large datasets, and to include features like automated backup and report printing in future updates. Overall, the system is now more reliable, faster, and easier to use helping village officials manage records more efficiently and accurately